

Root Finding Problems

Many problems in Science and Engineering are expressed as:

Given a continuous function $f(x)$,
find the value p such that $f(p) = 0$

These problems are called root finding problems.

Solution Methods

Several ways to solve nonlinear equations are possible:

- Analytical Solutions
 - Possible for special equations only
- Graphical Solutions
 - Useful for providing initial guesses for other methods
- Numerical Solutions
 - Open methods
 - Bracketing methods

Numerical Methods

Many methods are available to solve nonlinear equations:

- Bisection Method → bracketing method
- Newton's Method → open method
- Secant Method → open method
- False position Method (bracketing method)
- Fixed point iterations (open method)
- Muller's Method
- Bairstow's Method
-

Bracketing Methods

- In bracketing methods, the method starts with an interval that contains the root and a procedure is used to obtain a smaller interval containing the root.
- Examples of bracketing methods:
 - Bisection method
 - False position method (Regula-Falsi)

Open Methods

- ❑ In the open methods, the method starts with one or more initial guess points. In each iteration, a new guess of the root is obtained.
- ❑ Open methods are usually more efficient than bracketing methods.
- ❑ They may not converge to a root.

Outline

① Errors

② Root Finding Problem

③ Root Finding Methods

Bisection Method

Newton-Raphson's Method

Secant Method

The method of False Position (also called Regula Falsi)

④ Applications of Root Finding

Bisection Algorithm

Assumptions:

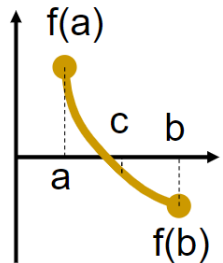
- $f(x)$ is continuous on $[a, b]$
- $f(a) f(b) < 0$

Algorithm:

Loop

1. Compute the mid point $c = (a+b)/2$
2. Evaluate $f(c)$
3. If $f(a) f(c) < 0$ then new interval $[a, c]$
If $f(a) f(c) > 0$ then new interval $[c, b]$

End loop



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Newton-Raphson's Method

The Eq. of tangent line passing through $(x_0, f(x_0))$ is

$$f(x) - f(x_0) = \underbrace{f'(x_0)}_m (x - x_0)$$

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Thus

$$-f(x_0) = f'(x_0) (x_1 - x_0)$$

Simplification gives

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

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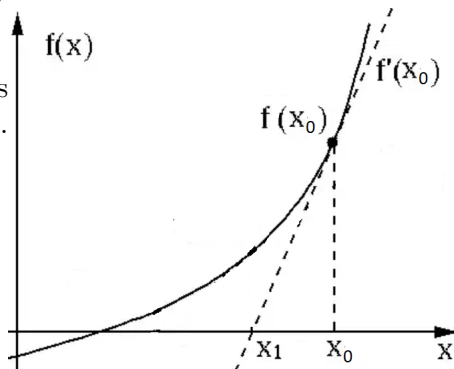
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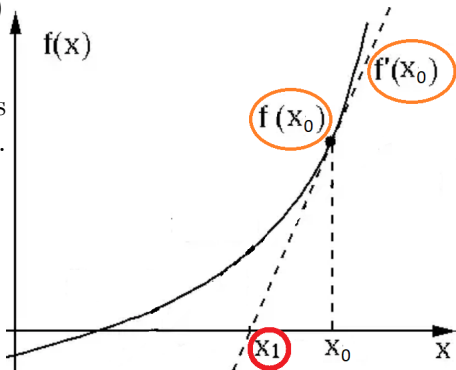
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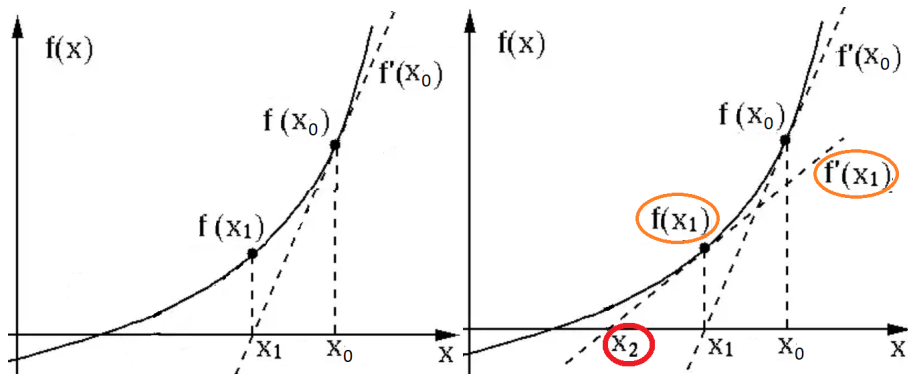
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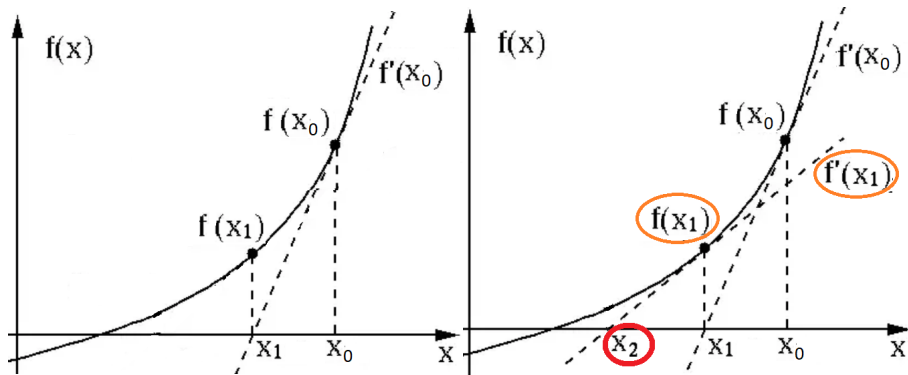
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$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}, \quad \text{provided } f'(x_1) \neq 0.$$



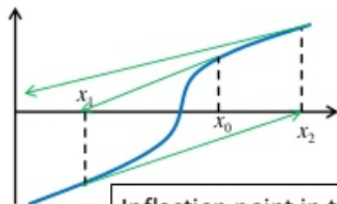
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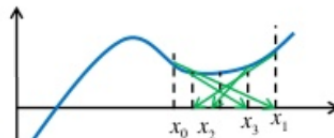


$$x_k = x_{k-1} - \frac{f(x_{k-1})}{f'(x_{k-1})}, \quad \text{provided } f'(x_{k-1}) \neq 0.$$

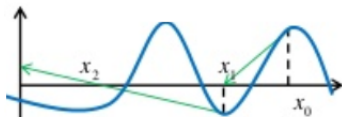
Pitfall of NR method



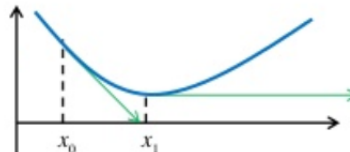
Inflection point in the vicinity of a root
> divergence



Oscillations around a maximum/minimum
> no convergence



Initial guess close to one root
jumps to another root
> solution jumps away



A zero slope is encountered
> Solution shoots-off

Determine the positive root of $f(x) = x^{10} - 1$ using the Newton-Raphson method and an initial guess of $x = 0.5$.

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Secant Method

The equation of Secant line for the function $f(x)$ passing through $(x_0, f(x_0))$ and $(x_1, f(x_1))$ is given by

$$\frac{f(x) - f(x_1)}{x - x_1} = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

The secant line intersect the x -axis at $x = x_2$ assuming $f(x_2) \simeq 0$ gives

$$\begin{aligned}\frac{0 - f(x_1)}{x_2 - x_1} &= \frac{f(x_1) - f(x_0)}{x_1 - x_0} \\ \frac{x_2 - x_1}{-f(x_1)} &= \frac{x_1 - x_0}{f(x_1) - f(x_0)}\end{aligned}$$

$$x_2 = x_1 - \frac{(x_1 - x_0) f(x_1)}{f(x_1) - f(x_0)} \quad \text{and in general,}$$

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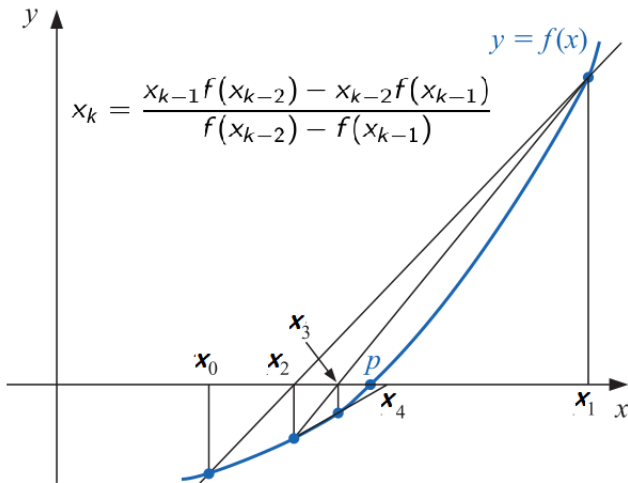
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$$x_k = x_{k-1} - \frac{(x_{k-2} - x_{k-1}) f(x_{k-1})}{f(x_{k-2}) - f(x_{k-1})} \quad \text{or} \quad x_k = \frac{x_{k-1} f(x_{k-2}) - x_{k-2} f(x_{k-1})}{f(x_{k-2}) - f(x_{k-1})}$$

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generates approximations in the same manner as the Secant method, but it includes a test to ensure that the root is always bracketed between successive iterations.

First choose initial approximations x_0 and x_1 with

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To decide which secant line to use to compute x_3 ,

- (a) If $f(x_2)f(x_1) < 0$, then x_1 and x_2 bracket a root. Choose x_3 as the x -intercept of the line joining $(x_1, f(x_1))$ and $(x_2, f(x_2))$.

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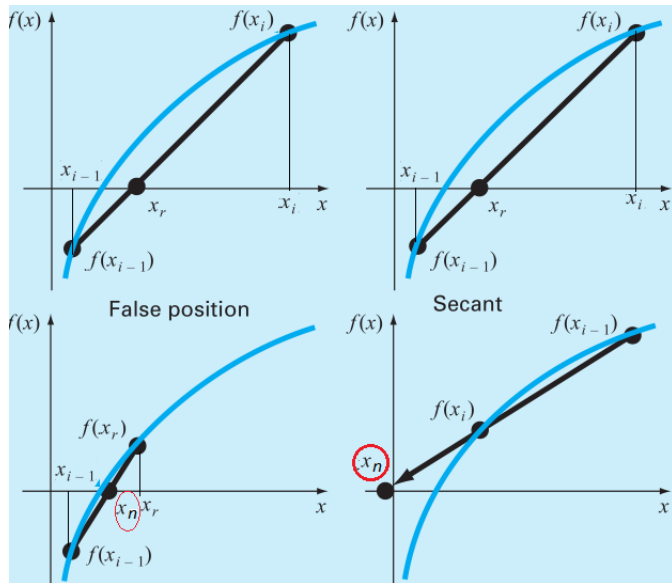
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- (b) If not, choose x_3 as the x -intercept of the line joining $(x_0, f(x_0))$ and $(x_2, f(x_2))$, and then interchange the indices on x_0 and x_1 .



Find a root for $\cos x - x + 2 = 0$ when $x_0 = 1$ and $x_1 = 2$

$$x_k = \frac{x_{k-1}f(x_{k-2}) - x_{k-2}f(x_{k-1})}{f(x_{k-2}) - f(x_{k-1})}$$

$$f(x_0) = 1.5403, f(x_1) = -0.41615$$

Secant Method:

Method of False Position:

$$x_2 = \frac{x_1 f(x_0) - x_0 f(x_1)}{f(x_0) - f(x_1)}$$

$$x_2 = 1.78729$$

$$x_3 = \frac{x_2 f(x_1) - x_1 f(x_2)}{f(x_1) - f(x_2)}$$

$$x_3 = 1.78621$$

$$x_4 = \frac{x_3 f(x_2) - x_2 f(x_3)}{f(x_2) - f(x_3)}$$

$$x_4 = 1.78623$$

$$x_2 = \frac{x_1 f(x_0) - x_0 f(x_1)}{f(x_0) - f(x_1)}$$

$$x_2 = 1.78729, f(x_2) \cdot f(x_0) < 0$$

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$$x_4 = \frac{x_3 f(x_2) - x_2 f(x_3)}{f(x_2) - f(x_3)}$$

$$x_4 = 1.78623$$

Find a root of the equation

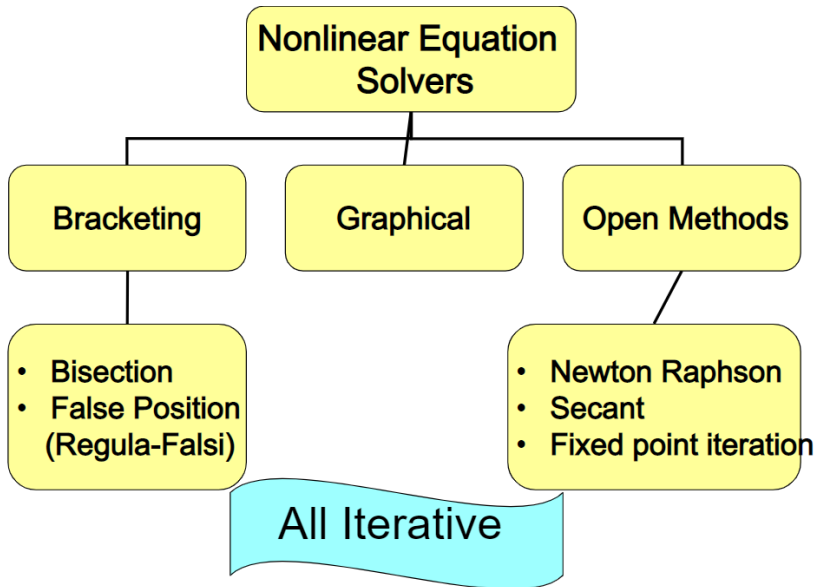
$$x = \cos x, \quad \left[0.5, \frac{\pi}{4}\right]$$

	False Position	Secant	Newton
n	p_n	p_n	p_n
0	0.5	0.5	0.7853981635
1	0.7853981635	0.7853981635	0.7395361337
2	0.7363841388	0.7363841388	0.7390851781
3	0.7390581392	0.7390581392	0.7390851332
4	0.7390848638	0.7390851493	0.7390851332
5	0.7390851305	0.7390851332	
6	0.7390851332		

① $4x = \sin x - e^x$

② $\cos x - xe^x$

③ $x^3 + 4x^2 - 10$



Applications of Root Finding

Electrical Engineering:

Find **resistance, R** , of a circuit such that the charge reaches q at specified time t

$$f(R) = e^{-Rt/2L} \cos \left[\left(\sqrt{\frac{1}{LC} - \left(\frac{R}{2L} \right)^2} \right) t \right] - \frac{q}{q_0} = 0$$

L = inductance,
 q_0 = initial charge

C = capacitance,

Applications of Root Finding

Civil Engineering:

Find **horizontal component of tension, H** , in a cable that passes through y_0 and (x, y)

$$f(H) = \frac{H}{w} \left[\cosh\left(\frac{wx}{H}\right) - 1 \right] + y_0 - y = 0$$

w = weight per unit length of cable

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